



Researchers build
1000 core processor
<http://dgit.in/1000cores>

Looking back at you

>>After years of you looking at the web, now the web is ready to look back at you> by Koustav Ray

Introduction

Eye and gaze tracking have been the domains of high end technology for a long time. But in the first week of June, computer scientists from Brown University have released a javascript library titled WebGazer developed by them that brings eye tracking within the reach of anyone with a website. Using the webcam of the computer or front cameras on mobile phones, this could be used by website owners to understand their users' areas of interest better and provide them content aligned with those interest areas in real time.

The eye tracking model is based on self calibration by analyzing the pattern of where the user is looking when he clicks a particular location on the website. The entire development has two main parts - a pupil detector that can be used with any eye detection library and a gaze estimator that uses regression analysis based on user interactions. This allows the software's self calibration to go on while the user is browsing the web. The images collected by the webcam during such browsing sessions can be analyzed to store what the user's eyes look like. After this, similar interactions will also be stored and the two will be compared to understand the relationship between eye features with respect to on-screen gaze location. This will allow the gaze to be tracked even when the user is not interacting.

That being said, that is not the only way in which WebGazer tracks the gaze of the users. The second one treats the eye as a multi-dimensional feature vector and combines user interactions with regularized linear regression. This goes beyond the usage of clicks for tracking, it also takes into account cursor movement and eye-cursor coordination delay as vital inputs to estimate the gaze accurately.

The relative simplicity of WebGazer might be its strongest suit. It has no 3D capabilities that render it light and easy to implement into any browser and website. It makes up for the loss of accuracy there by constant calibration. Along with that, the constant calibration removes the need for any initial calibration session and lets the user be free to move during any tracking. Pretty soon, it might become internet norm to have websites tracking your gaze, quite like the crawlers of today

Getting WebGazer:

The entire WebGazer library is a simple javascript file available for download on their website.

To get it, simply go to <https://WebGazer.cs.brown.edu/#download> and download the file onto your machine. When using the library on a webpage, do remember to include it by including it as follows:

```
<script src="WebGazer.js" type="text/javascript">
```

*Coding Matters



*"Google and Udacity launch Android course"

>> Android basics nanodegree launched for beginners with zero coding experience.

<http://dgit.in/UdaAndroid>



*"RedEye gives your phone 24x7 vision"

>> New technology to give computers intelligent vision that remembers specific things.

<http://dgit.in/RdEye>



*"ECMAScript 2016 approved"

>> Also known as JavaScript 1.9 and ES7, yearly release of ECMAScript with couple of new features .

<http://dgit.in/ECMA2016>